

Target

Improving educational outcomes including the ethical, responsible, and inclusive use of A.I. for students.

My Brief Summary

This project addresses persistent barriers in higher education: college readiness, academic support, and disengagement in gateway mathematics courses. Many incoming students lack confidence, foundational skills, or consistent academic support during the critical onboarding phase. To address this, we implemented Study Fetch as an AI-powered tutoring system that provides students with on-demand, personalized academic support specifically aligned to course content.

Faculty integrate the platform into classroom instruction to enhance engagement and course activities to reinforce foundational concepts. Prior to class and at the beginning of class, students engage with the platform through AI agents acting as tutors to course materials and learning outcomes. These agents use interactive questioning, adaptive feedback, and step-by-step problem-solving strategies to meet students at their level of understanding, support their growth, and facilitate their demonstration of the learning outcomes. Even after class, these agents provide additional support to students when they need it and in a manner that they can understand. The platform gives underprepared and underperforming students access to the scaffolding they need to prepare for future content and achieve success.

Initial outcomes indicate increased student engagement with course materials, improved confidence in problem-solving, and more consistent participation throughout the course. Students who actively use AI tutoring tools demonstrate stronger persistence in completing assignments and are more likely to seek help proactively. Importantly, this approach expands access to academic support for students who may not utilize traditional tutoring services.

This model is scalable across disciplines and institutions. It leverages the adaptability of a large language model built on instructor-provided course materials and objectives. It embeds instructor-guided support directly into the student learning experience. It even supports instructors as they design, improve, and sustain the platform. Wherever students and faculty need to support learning new content directed by learning outcomes, this model can be used.

Overall, this project demonstrates how AI can be used not to replace instruction, but to extend it in ways that are practical, inclusive, and impactful.

AI-Enhanced Summary

Project Overview

This project addresses persistent barriers in higher education: college readiness, access to academic support, and disengagement in gateway mathematics courses. Many incoming students lack confidence, foundational skills, and consistent support during the critical onboarding phase. To address this, we

implemented Study Fetch as an AI-powered tutoring system that provides on-demand, personalized academic support aligned with course content and learning outcomes.

Description of AI Use

Faculty integrate the platform directly into instruction to enhance engagement and reinforce foundational concepts. Before and during class, students interact with AI agents that function as tutors, guiding them through course materials using interactive questioning, adaptive feedback, and step-by-step problem-solving. These agents meet students at their current level of understanding, support skill development, and help them demonstrate mastery of learning outcomes.

Outside of class, students continue to access support in real time, receiving explanations tailored to their needs. This approach provides underprepared and underperforming students with the scaffolding necessary to engage with new content and progress successfully.

Evidence of Impact

Initial outcomes indicate increased engagement with course materials, improved confidence in problem-solving, and more consistent participation throughout the course. Students who actively use AI tutoring tools demonstrate stronger persistence in completing assignments and are more likely to seek help proactively.

Student feedback reinforces these observations. One student shared that “it is easier to learn with Study Fetch... since it is giving step by step [support] as well as allowing for examples to be put to use,” highlighting the value of guided, applied learning. Another noted that the platform “helps me with learning new things and new tactics that I have never tried before,” pointing to increased exploration and skill development.

When asked which aspects of the course supported their learning most, one student responded, “Probably the Study Fetch assignments because it’s like a teacher talking with you and helping you out through the process.” This reflects the platform’s ability to provide accessible, personalized support that mirrors one-on-one instruction.

This model also expands access to academic support for students who may not utilize traditional tutoring services, helping to close gaps in preparedness and participation during the early stages of the course.

Ethical Considerations

Ethical implementation is central to this project, with a focus on transparency, accuracy, student agency, and equitable access. Students clearly understand when they are interacting with AI, as its use is confined to Study Fetch. The instructor models effective use and engages students in evaluating AI responses, using occasional errors as opportunities to reinforce verification and critical thinking.

AI support is aligned with instructor-designed materials and closely monitored. Students are encouraged to question and verify responses, while the instructor actively refines activities to ensure accuracy and consistency with learning objectives.

Student agency is preserved through intentional design. An evaluator agent prioritizes mastery by rewarding less scaffolded work, promoting independent problem-solving. A syntax-support agent further reduces access barriers for students unfamiliar with mathematical notation.

Privacy is addressed through an institutional agreement that limits the use of student data to the platform. The system also supports inclusivity by allowing students to request explanations tailored to their interests and learning preferences.

Overall, AI is positioned as a transparent, instructor-guided support that enhances learning while maintaining student responsibility and equitable access.

Scalability

This model is highly portable, adaptable, and sustainable across courses, disciplines, and institutions. Implementation is grounded in standard instructional materials—Content Outlines, Learning Outcomes, and Pedagogical Frameworks—allowing instructors to adopt the model without extensive technical expertise. Because instructors can use the platform itself to learn its functionality, onboarding is efficient, and ongoing refinement becomes the primary focus.

The approach extends beyond mathematics. AI agents are guided by pedagogical frameworks rather than discipline-specific constraints, meaning only the content and learning outcomes need to change for use in other subject areas. Core elements—such as guided questioning, adaptive feedback, and mastery-based evaluation—are fully transferable.

Faculty adoption is supported through reusable templates and structured agent designs. Once established, activities can be created efficiently, with initial setup requiring only modest time investment. Professional learning communities or collaboration with experienced faculty or instructional designers can further streamline implementation and support continuous improvement. While workload decreases over time, periodic review ensures content accuracy and alignment with course goals.

Although this model was implemented using Study Fetch, it is not platform dependent. Similar approaches could be adapted using other AI tools such as Microsoft Copilot, Google Gemini, or ChatGPT. Platform features such as interaction logs and analytics further support scalability by enabling instructors to monitor usage, address academic integrity concerns, and refine implementation.

The model is sustainable over time, with minimal barriers to updating or expanding content. As instructors revise the platform with new materials and share practices, the system becomes increasingly efficient and impactful, supporting broader adoption at the course, program, or institutional level.

Conclusion

This project demonstrates a practical and effective model for integrating AI into gateway mathematics in ways that directly support student learning and persistence. By embedding Study Fetch into instruction, the approach extends academic support beyond the classroom, meeting students where they are and guiding them toward mastery. Grounded in clear pedagogy, supported by intentional design, and strengthened through ethical implementation, this model positions AI as a tool that amplifies high-quality teaching and just-in-time support. Its adaptability, sustainability, and focus on student success make it a compelling framework for broader adoption across courses, disciplines, and institutions.